# VIDYA JYOTHI INSTITUTE OF TECHNOLOGY

**AN AUTONOMOUS INSTITUTION**

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## DEPARTMENT OF INFORMATION TECHNOLOGY


# LAB PROJECT REPORT

| | | |
|---|---|---|
| **Roll No** | : | **24911A1243** |
| **Name** | : | **Neerati Rupasri** |
| **Project Title** | | **Online Examination System** |
| **Course Name** | : | **Object Oriented Programming through Java Lab** |
| **Course Code** | : | **A224592** |
| **Academic Year** | : | **2025-26** |
| **Regulations** | : | **R22** |
| **Year and Semester** | : | **II B.Tech II Semester** |

# CERTIFICATE

This is to certify that the Object Oriented Programming through Java Lab project report entitled “Online Examination System” submitted by **Neerati Rupasri (24911A1243)** to Vidya Jyothi Institute of Technology (An Autonomous Institution), Hyderabad, of B. Tech. II year, II Semester in Information Technology a Bonafide record of Lab project work carried out by them under my supervision.

| **Signature of Faculty** | **Signature of Head of the Department** |
|---|---|
| Mr. B Eswar Babu | Dr. A. Obulesu |
| Associate Professor | Professor |

## Abstract

The Online Examination System is a Java-based desktop application developed to automate the process of conducting examinations and evaluating student performance. Traditional examination methods consume significant time for paper preparation, manual evaluation, and result processing. This project provides a computerized solution that allows users to attend multiple-choice examinations online with automatic evaluation and score generation.

The main objective of the project is to create a secure, efficient, and user-friendly examination platform using Java technologies. The system includes features such as user login authentication, multiple-choice question management, timer functionality using multithreading, automatic answer evaluation, and instant score display. The application uses Java Swing for graphical user interface development, JDBC for database connectivity, and MySQL for data storage.

The project also demonstrates important Real-Time Java concepts such as Threads, Exception Handling, Object-Oriented Programming, GUI programming, and Database Connectivity. The timer module ensures that exams are completed within the specified duration, while automatic evaluation reduces human effort and minimizes errors.

The expected outcome of the project is an efficient online examination platform that improves examination management, reduces paperwork, saves time, and provides quick and accurate results. The system can be further enhanced with advanced security features, web deployment, and cloud integration for large-scale usage.

# Table of Contents

# Chapter 1 – Introduction

## 1.1 Project Overview

The Online Examination System is designed to conduct examinations digitally through a computer-based application. The system allows students to log in, attend multiple-choice tests, and receive instant results after submission. Administrators can add questions, manage exams, and monitor student performance.

The project reduces manual work involved in conducting examinations and evaluating answer sheets. It provides a fast, secure, and reliable method for online testing. The timer functionality ensures fair examination practices by limiting the exam duration.

Real-World Applications

- Educational institutions
- Online certification platforms
- Recruitment examinations
- Competitive exam preparation systems

## 1.2 Objectives

- To automate the examination process
- To reduce manual evaluation errors
- To provide quick result generation
- To improve efficiency and accuracy
- To implement Java real-time concepts practically
- To maintain secure student records

## 1.3 Scope of the Project

The system is mainly used by:

- Students
- Teachers
- Administrators

**Future Scalability**

- Web-based deployment
- Mobile application support
- Cloud database integration
- AI-based performance analysis

# Chapter 2 – System Analysis

## 2.1 Functional Requirements

The system should perform the following functions:

- User login authentication
- Conduct online examinations
- Display multiple-choice questions
- Timer management using threads
- Automatic evaluation
- Score generation
- Database storage of student records
- Admin question management

## 2.2 Non-Functional Requirements

Security

- Password protection
- Authorized access only

Performance

- Fast response during exams
- Efficient database connectivity

Reliability

- Accurate result generation

- Stable system execution

Usability

- Simple graphical user interface
- Easy navigation

## 2.3 Software Requirements

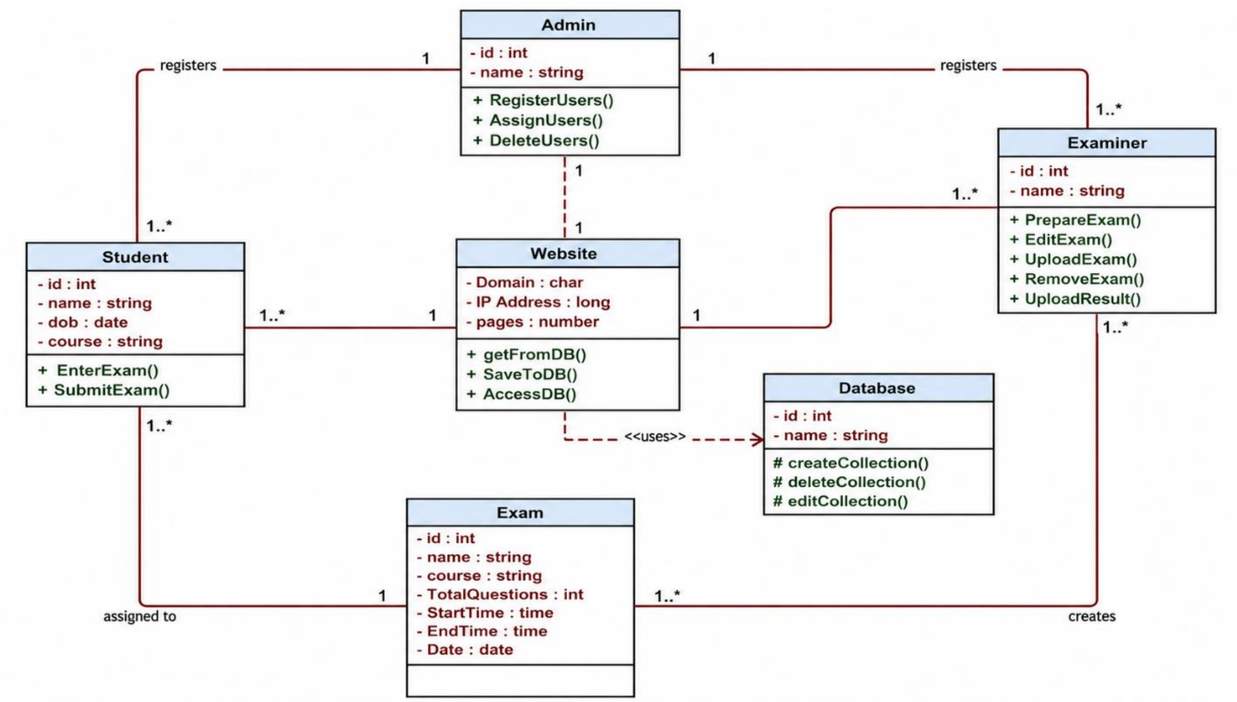
| Requirement | Details |
|---|---|
| OS | Windows/Linux |
| Language | Java |
| IDE | NetBeans/Eclipse |
| Database | MySQL |
| JDK | JDK 8 or above |

## 2.4 Hardware Requirements

| Requirement | Details |
|---|---|
| RAM | 4 GB |
| Processor | Intel i3 or above |
| Storage | 500 GB |

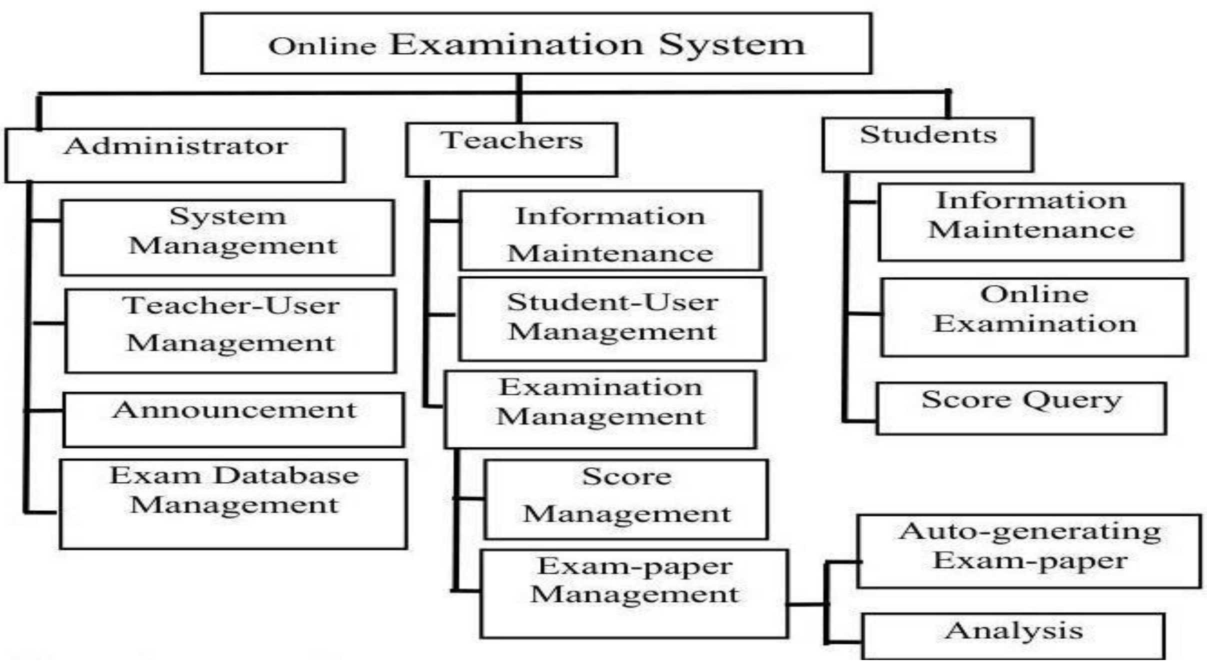
# Chapter 3 – System Design

## 3.1 Architecture Diagram

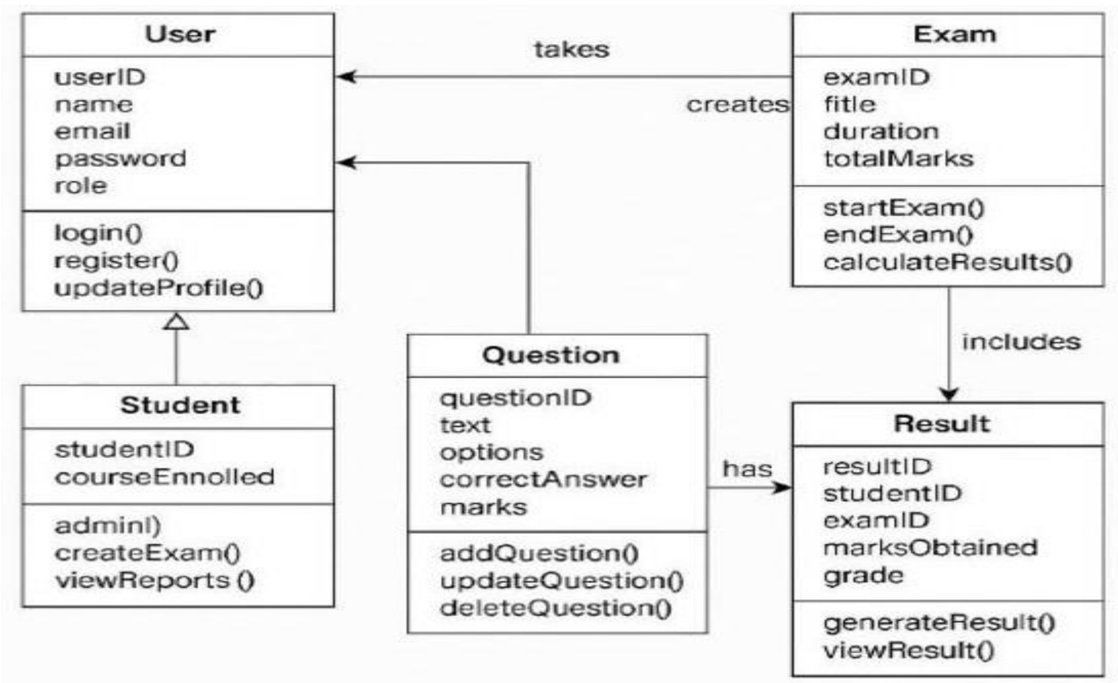



## 3.2 UML Diagrams

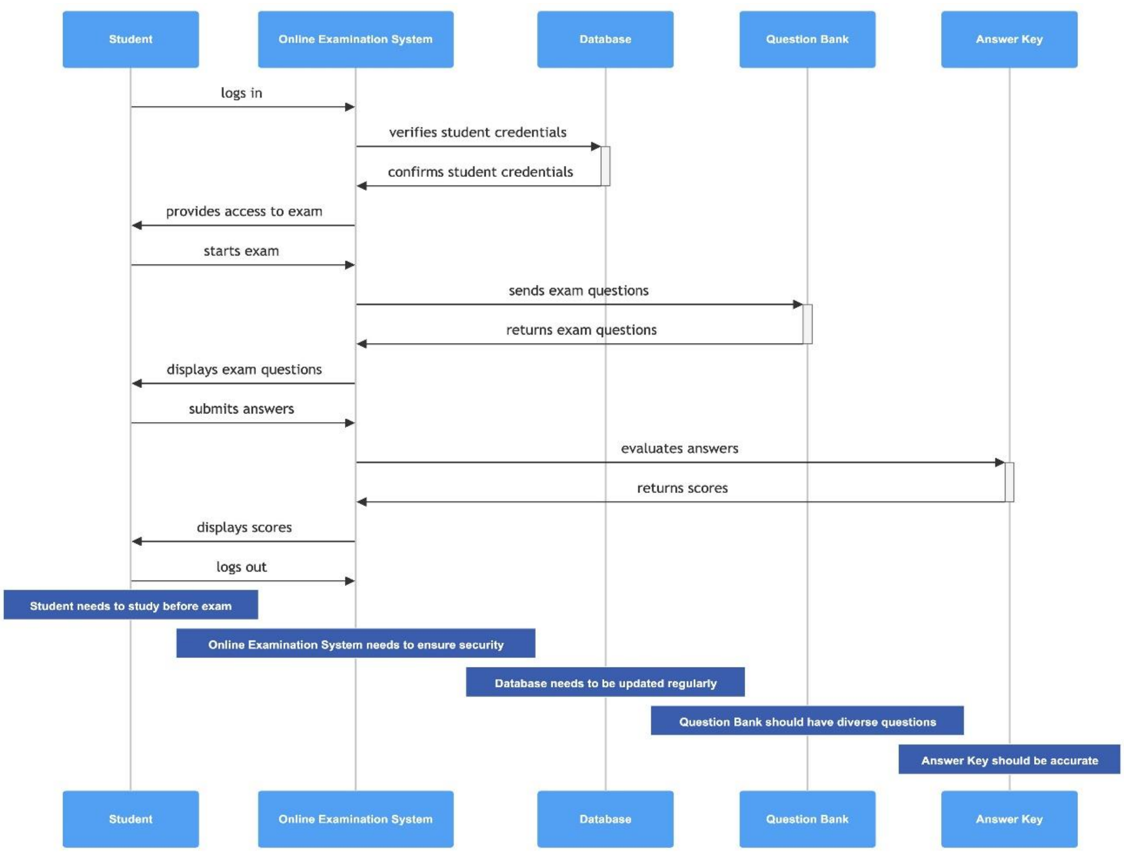
- Use Case Diagram



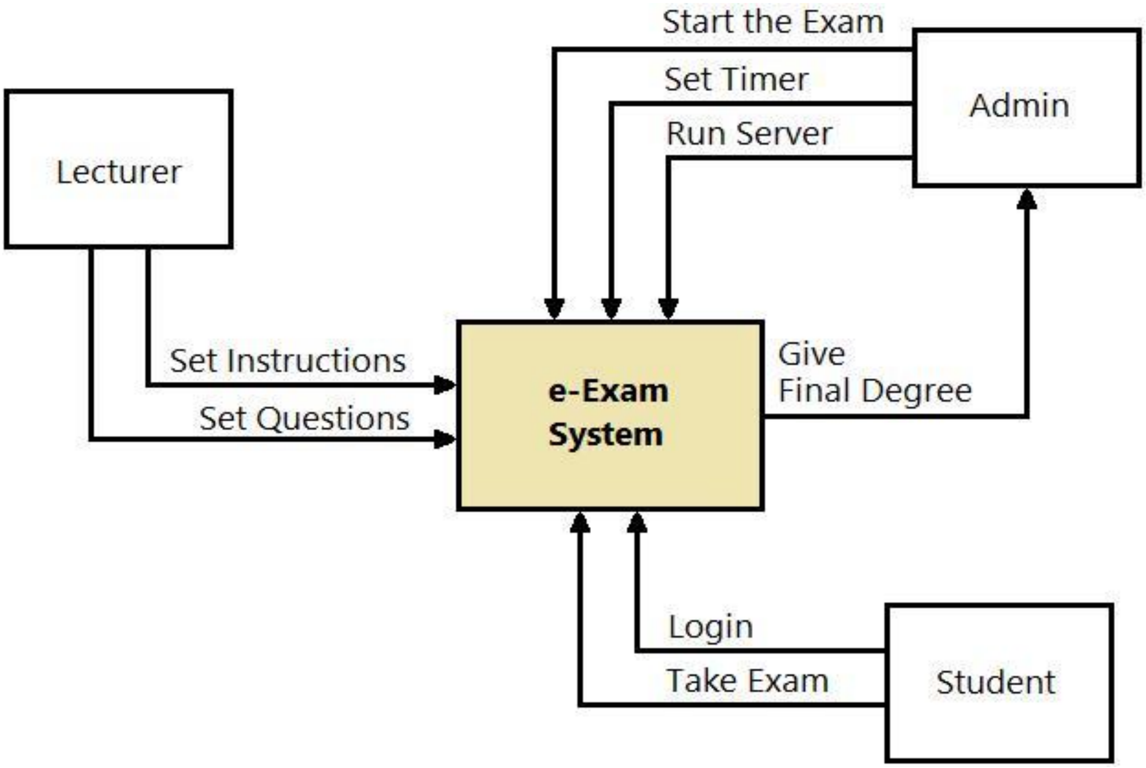
- Class Diagram




- Sequence Diagram



- Activity Diagram




**3.3 Database Design**

| **Table Name** | **Purpose** |
|---|---|
| users | Store login details |
| questions | Store examination questions |
| results | Store student scores |

# Chapter 4 – Implementation

## 4.1 Modules Description

| Module | Description |
|---|---|
| Login Module | User authentication |
| Exam Module | Conduct online examination |
| Timer Module | Manage exam duration using threads |
| Evaluation Module | Automatic answer checking |
| Report Module | Generate scores and reports |

## 4.2 Important Java Concepts Used

The project uses the following Java concepts:

- Classes and Objects
- Inheritance
- Polymorphism
- Exception Handling
- Multithreading
- Collections
- JDBC
- AWT/Swing GUI

### 4.3 Code Snippets

Database Connection Code:

```
import java.sql.Connection;

import java.sql.DriverManager;

public class DBConnection {

    Connection con;

    public Connection getConnection() {

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            con = DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/examdb",

                "root",

                "password");

        } catch(Exception e) {

            System.out.println(e);

        }

        return con;

    }

}
```

Timer Thread Code:

```
class TimerThread extends Thread {
    int time = 60;
    public void run() {
        try {
            while(time > 0) {
                System.out.println("Time Left: " + time);
                time--;
                Thread.sleep(1000);
            }
        } catch(Exception e) {
            System.out.println(e);
        }
    }
}
```

Login Validation Code:

```
if(username.equals("admin") && password.equals("1234")) {
    System.out.println("Login Successful");
} else {
    System.out.println("Invalid Credentials");
}
```

# Chapter 5 – Testing

## 5.1 Test Cases

| Test Case ID | Input | Expected Output | Result |
|---|---|---|---|
| TC01 | Valid Login | Login Success | Pass |
| TC02 | Invalid Password | Error Message | Pass |
| TC03 | Submit Correct Answers | Score Generated | Pass |
| TC04 | Timer Ends | Auto Submit | Pass |

## 5.2 Error Handling

The following exceptions are handled in the project:

- NumberFormatException
- SQLException
- NullPointerException
- FileNotFoundException

# Chapter 6 – Results and Screenshots

Figure 1 – Login Screen


Figure 2 – Main Menu

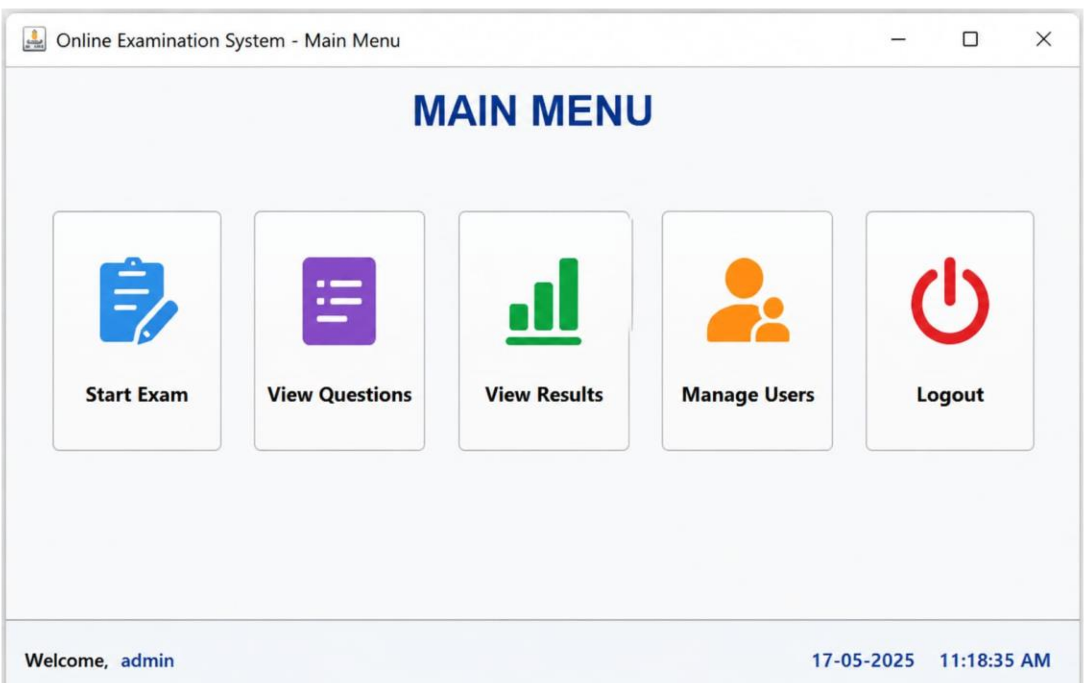


Figure 3 – Online Test Screen

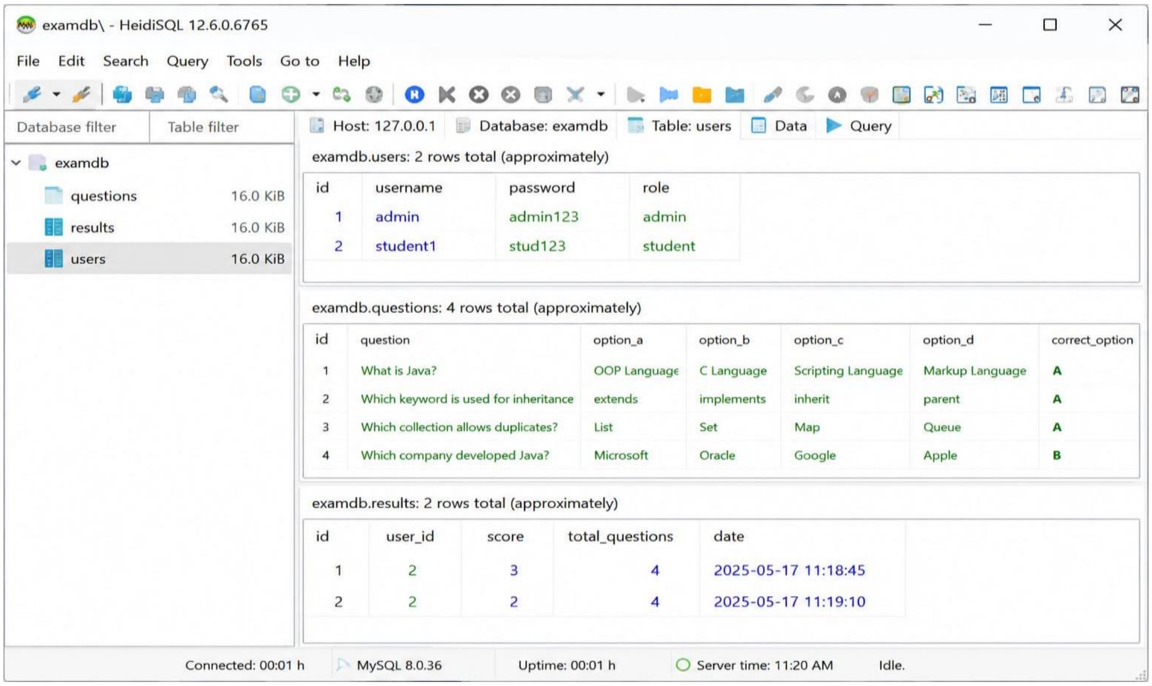



Figure 4 – Result Screen

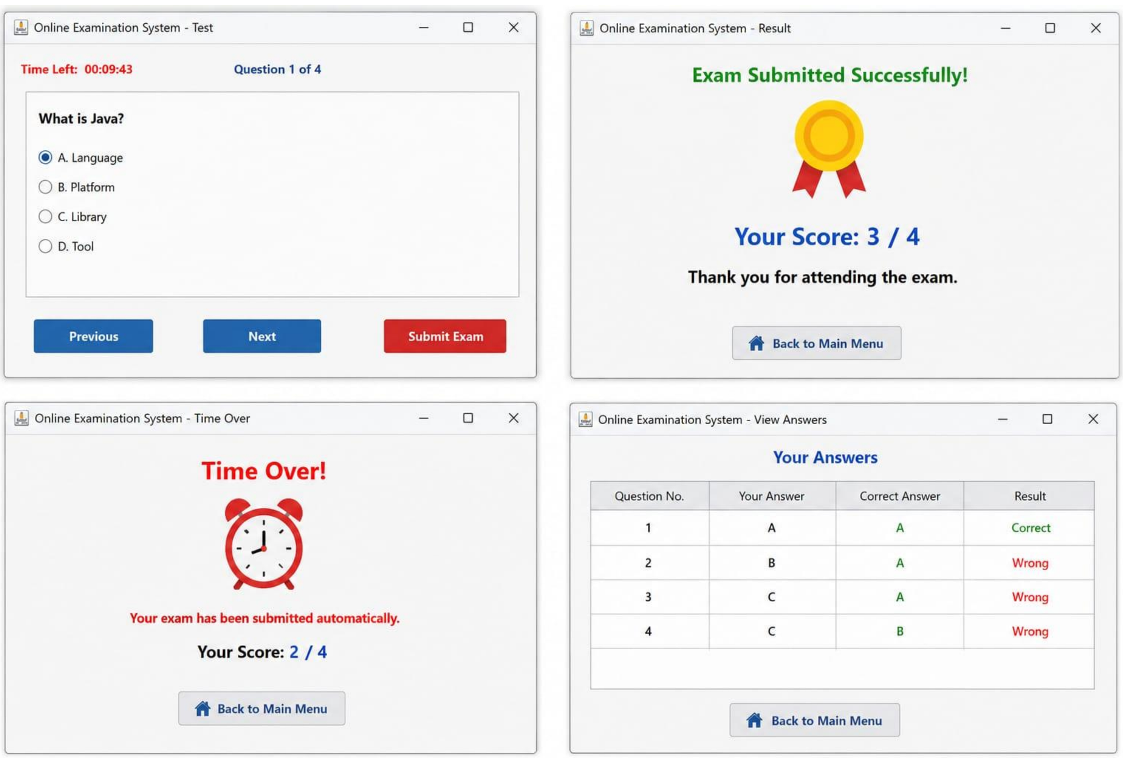
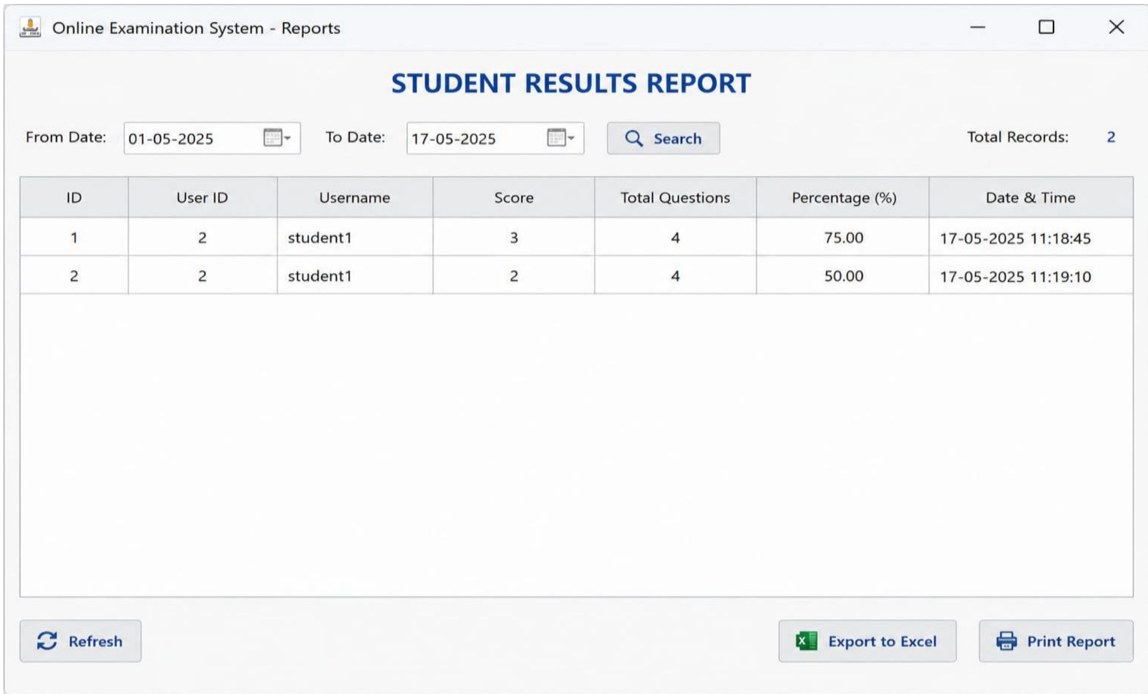


Figure 5 – Database Table



# Chapter 7 – Conclusion and Future Scope

## Conclusion

The Online Examination System was successfully developed using Java and MySQL technologies. The project automates the examination process and reduces manual effort involved in conducting tests and evaluating answer sheets. The system provides secure login authentication, automatic evaluation, and instant score generation.

This project helped in understanding important Real-Time Java concepts such as multithreading, exception handling, GUI development, and JDBC connectivity. The system improves efficiency, accuracy, and reliability in examination management.

## Future Enhancements

- Cloud database integration
- Mobile application support
- AI-based performance analysis
- Online payment integration
- Web-based deployment
- Biometric authentication

# References


1. Java: The Complete Reference
2. Thinking in Java
3. MySQL Workbench
4. Oracle Java Documentation